

Yufeng (Parra) Wang

MSE, Mechanical Engineering · University of Pennsylvania

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Research

Support-caster behavior on HAMR / CHUTNI

Independent study with Prof. Mark Yim · University of Pennsylvania · 2025–present

Integrated a passive split-spherical support caster and refined mounting interfaces for height, clearance, and contact. Worked on ROS 2 closed-loop tracking with Vicon, frame/yaw alignment, controller tuning, and motion logging. Developing a controlled comparison with conventional trailing casters.

Active spherical caster

Ongoing mechanism-design direction · 2026

Kinematic modeling and CAD for a standalone actively driven caster. Prototype fabrication and control implementation are next.

Education

University of Pennsylvania

MSE, Mechanical Engineering · August 2025–May 2027, expected · GPA 4.0/4.0

University of Pittsburgh

BSE, Mechanical Engineering · August 2023–May 2025 · GPA 3.895/4.0

Sichuan University

Mechanical engineering study · September 2021–July 2023

Selected projects

Quadrotor autonomy and local replanning

MEAM 620 · Spring 2026

3D occupancy-grid A*, waypoint simplification, trajectory generation, SE(3) control, visual-inertial estimation, and a receding-horizon local-replanning extension.

Autonomous competition robot

MEAM 5100 · Fall 2025

Mechanical design lead for chassis layout, drivetrain packaging, ramp-capable caster suspension, and sensor/actuator mounts. Contributed to wall following, Vive navigation, and target-interaction integration.

Autonomous cube-stacking robot arm

MEAM 5200 · Fall 2025

Camera-based detection, FK/IK, grasp tuning, gripper feedback, timed dynamic-block capture, and supporting potential-field/RRT planning work.

MEMS vibratory gyroscope

ESE / MEAM 5290 · Spring 2026

Team design study with comb-drive actuation, differential capacitive sensing, mode matching, and a proposed SOI/DRIE fabrication process.

Desktop tensile test machine

Mechanical Design 2 · University of Pittsburgh · Spring 2024

Gear design, lead-screw stress and buckling analysis, and spreadsheet-based stress, torque, and speed calculations.

Engineering experience

Chenzhi Technology

Engineer Assistant, Steering System Design · May–August 2024

Designed steering-system housings in CATIA V5, supported assembly and debugging, and adjusted rack-and-pinion clearance, motor placement, and belt tension. Worked with cast-component inspection, bearing comparisons, and interface specifications.

Tools and methods

Robotics: ROS 2, Gazebo, Vicon, rosbag2, PID and SE(3) control, kinematics, A*, RRT, VIO.

Mechanical / embedded: SolidWorks, CATIA V5, Arduino, ESP32, mechanical integration and prototyping.

Programming: Python, C++, MATLAB, Java.